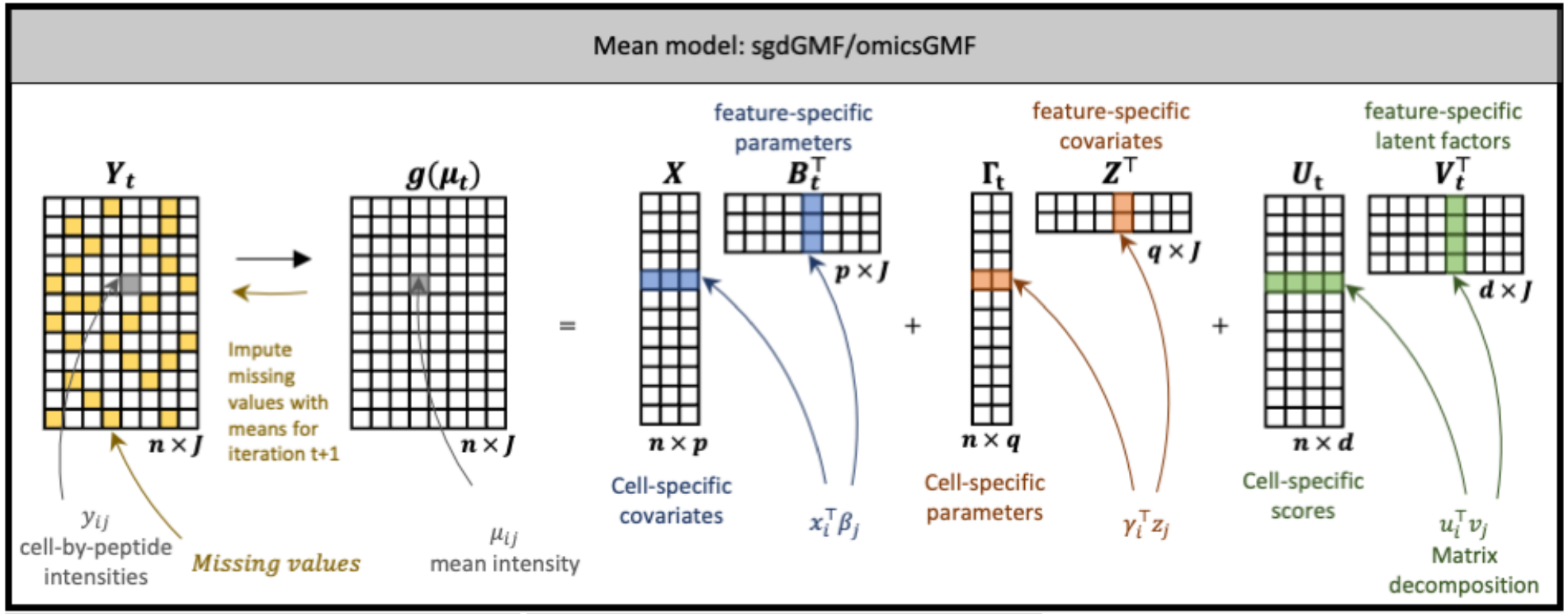
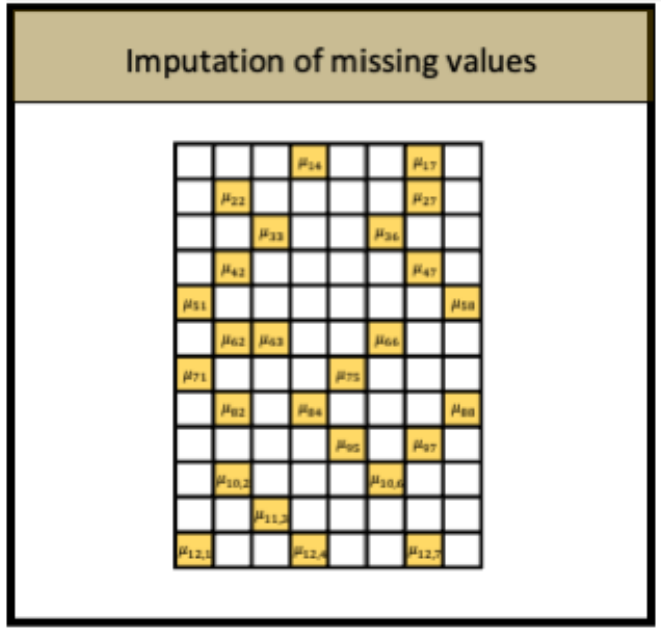
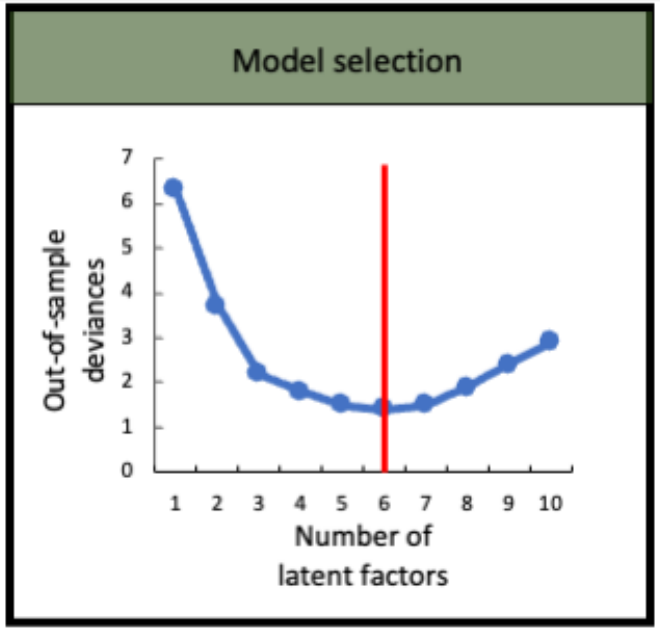
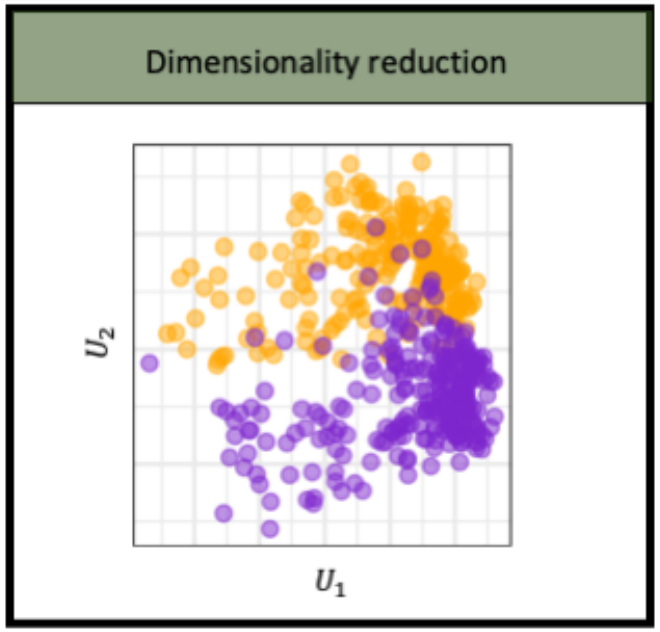
omicsGMF: dimensionality reduction, batch correction and imputation for (single cell) omics data

Alexandre Segers, Cristian Castiglione, Christophe Vanderaa, Lennart Martens, Davide Risso, Lieven Clement









Gen. matrix factorisation
- Exponential family

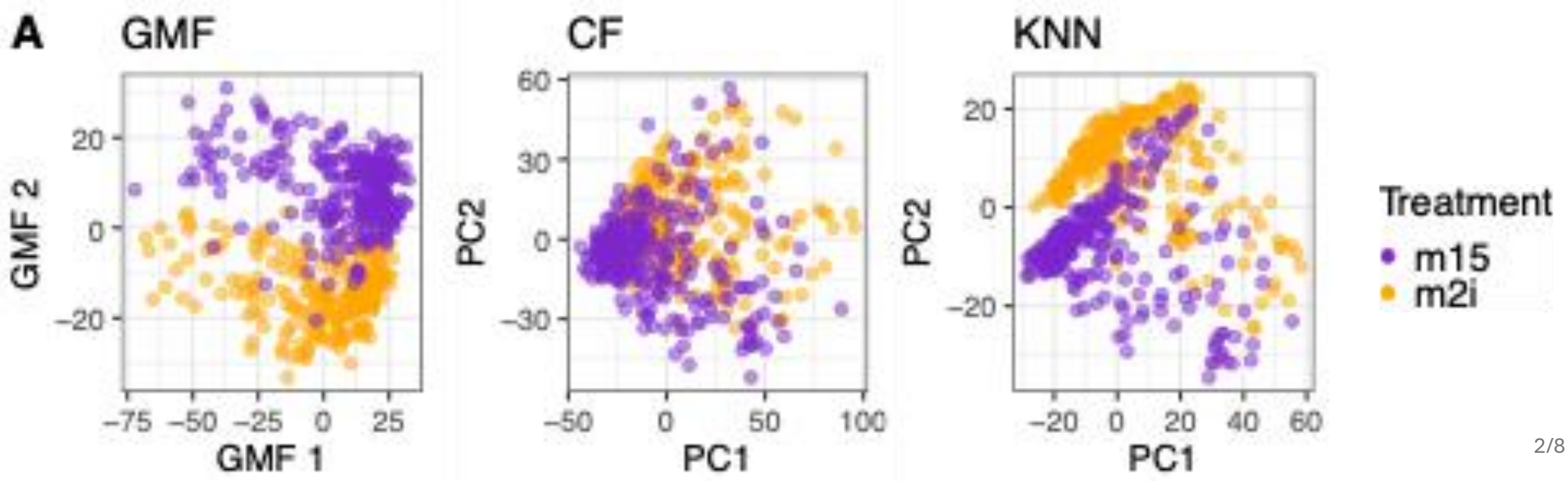
RNAseq
- NB or Poisson
- log-link
- sgdGMF preprint

Mass-spec. Proteomics
- Normal upon log-trans
- Identity link
- Focus of this talk & omicsGMF preprint

Implementation
- Fast C++
- SGD (Mini-blocks)
- Classes: QFeatures, Summarized & singleCell experiment

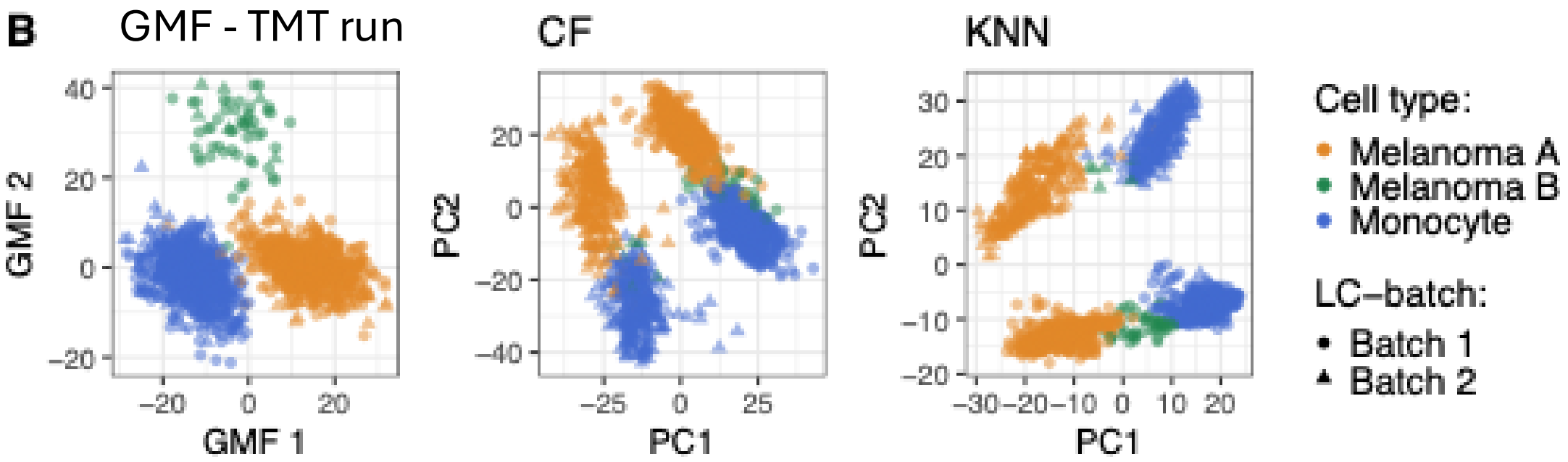
Petrosius Single Cell Proteomics (SCP) Data

- 525 cells treated with / without inhibitor (m2i/m15)
- Label-free → one cell per MS-run
- 4435 peptides, 58.5% missing values
- Competitors (PIMMS-CF, KNN, ...): imputation followed by PCA



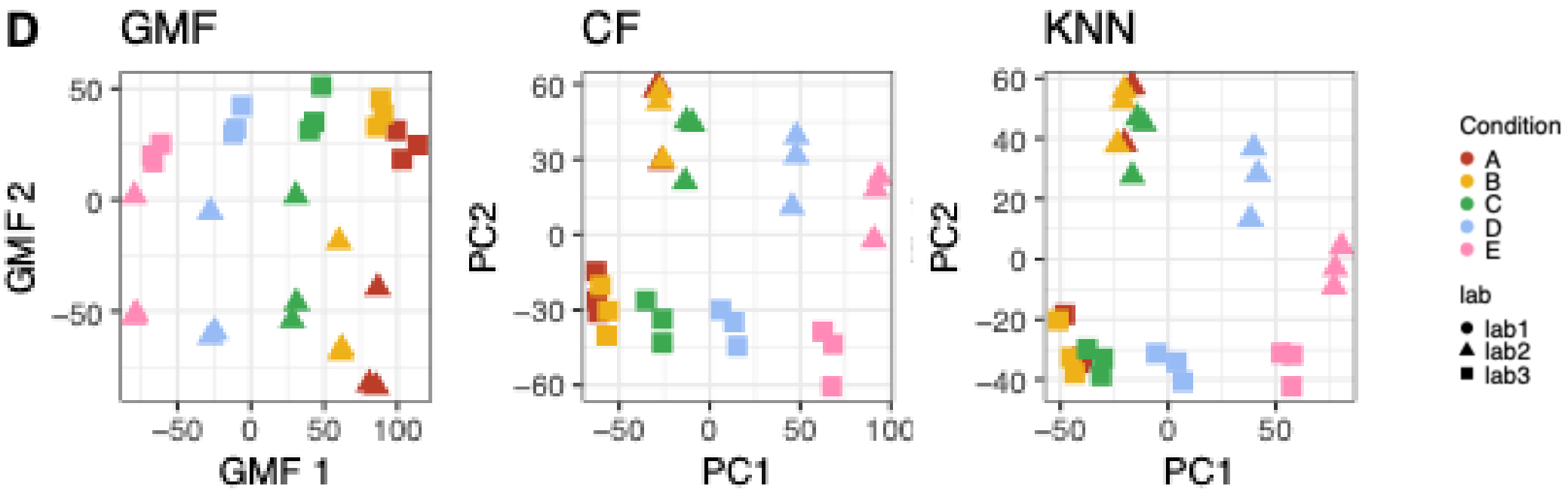
Leduc SCP Data

- 1508 cells from two cell lines: melanoma and monocytes
- TMT-multiplexing → up to 14 cells/MS-run → strong run effect
- Change LC-column → LC batch effect
- 4435 peptides, 61.8% missing values

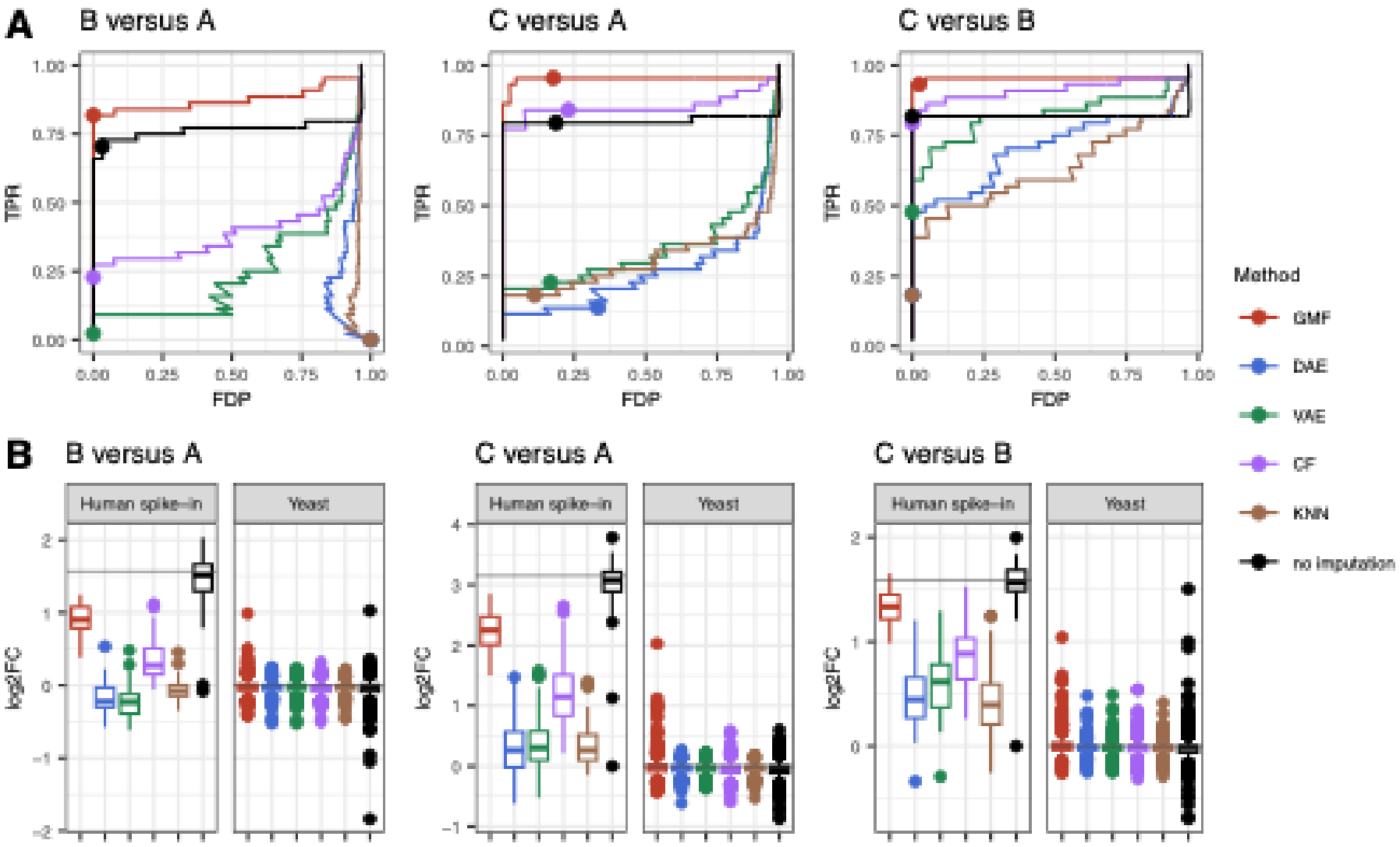


CPTAC Spike-in Bulk Proteomics Data

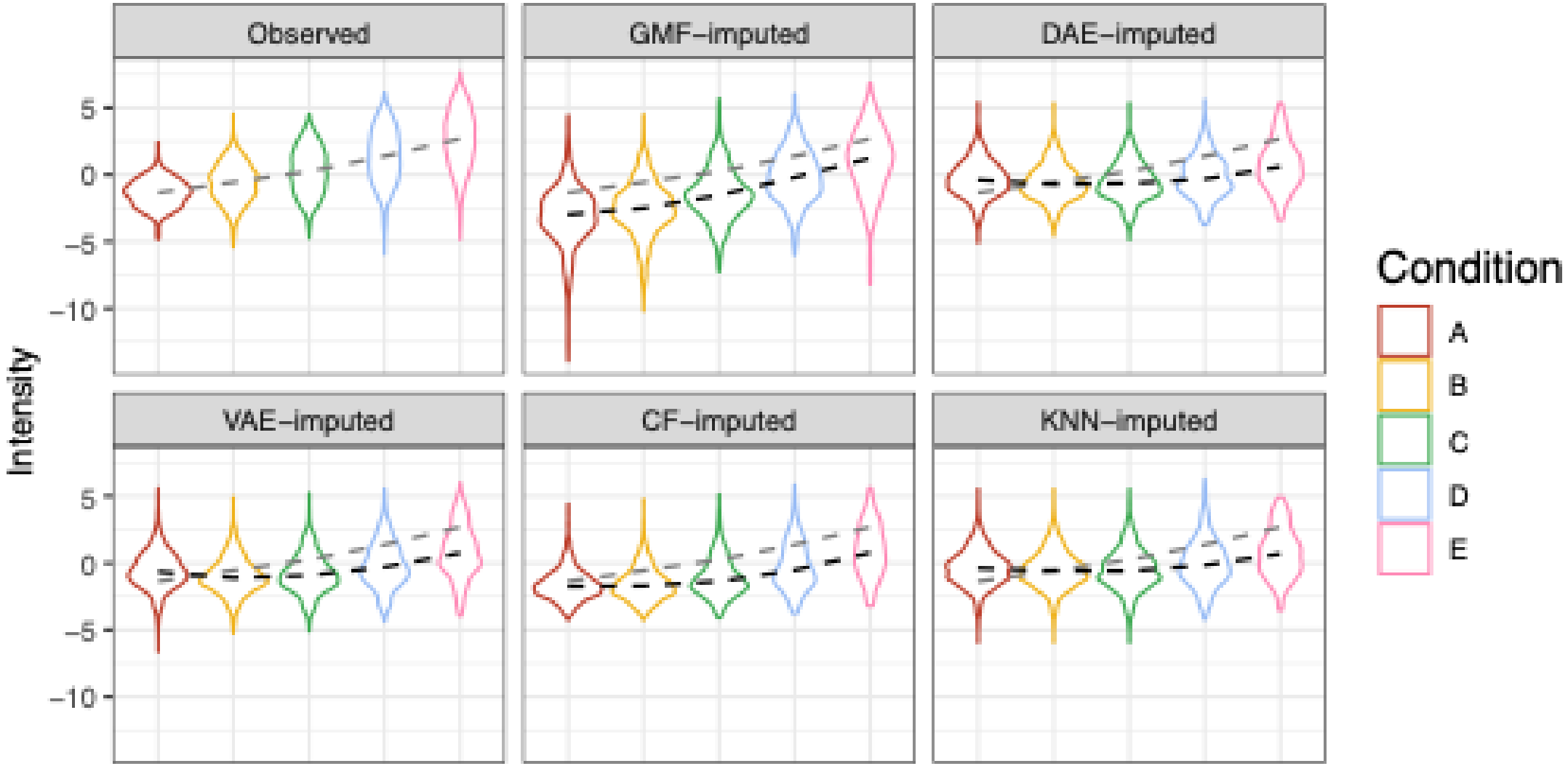
- Yeast background with 48 UPS proteins spiked in at 5 levels
- Analysed in triplicate in three labs → 45 samples
- 10105 peptides, 42.5% missing values



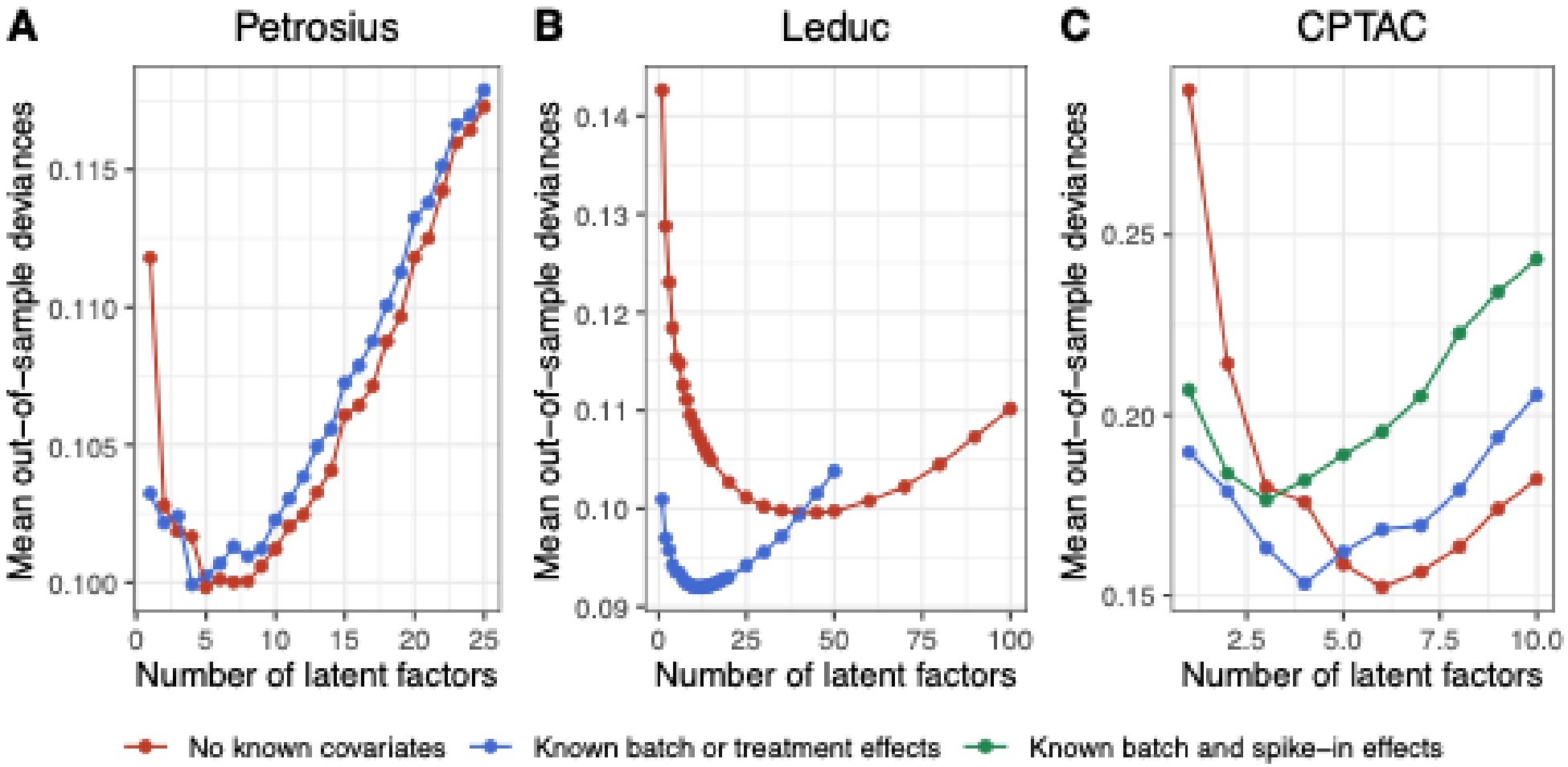
CPTAC Spike-in Differential Abundance Analysis



Imputation: Missing Spike-in Peptides



Model Selection via Cross-Validation



Take Home Messages

- Simultaneous dimensionality reduction, batch correction and imputation
- For bulk and single-cell proteomics (1) & transcriptomics (2)
- Model selection for known & # latent variables

→ Improves quality of dimensionality reduction & imputation in proteomics

Pre-prints

(1) https://www.biorxiv.org/content/10.1101/2025.03.24.644996v1
(2) https://arxiv.org/abs/2412.20509

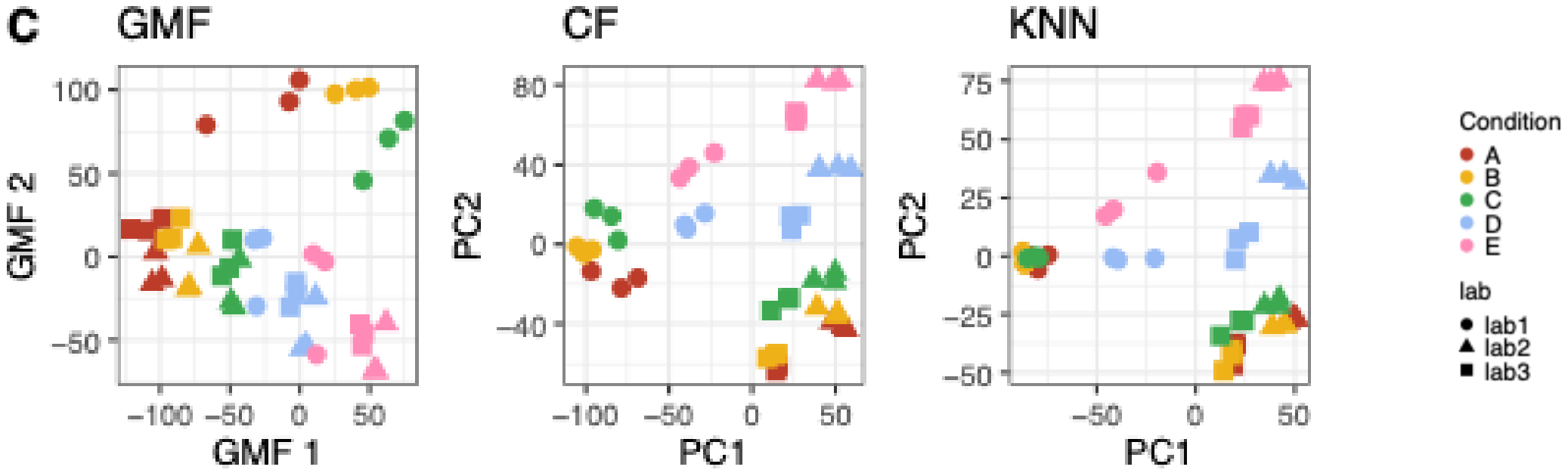
R/Bioconductor-packages

https://github.com/statOmics/omicsGMF (1,2)
https://cran.r-project.org/web/packages/sgdGMF

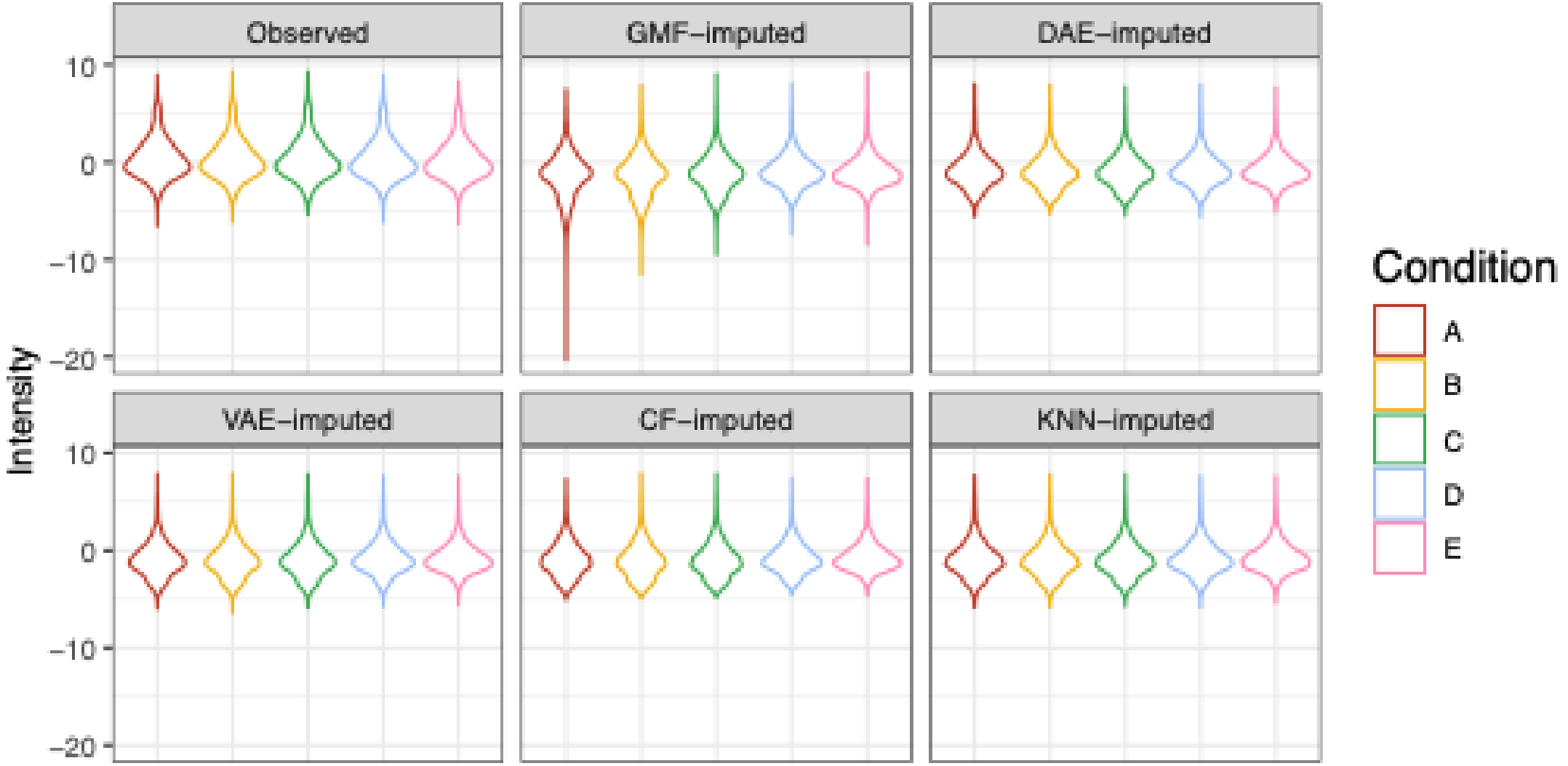
fwo MUR Università degli Studi di Padova VIB Universiteit Gent

CPTAC Spike-in Bulk Proteomics Data

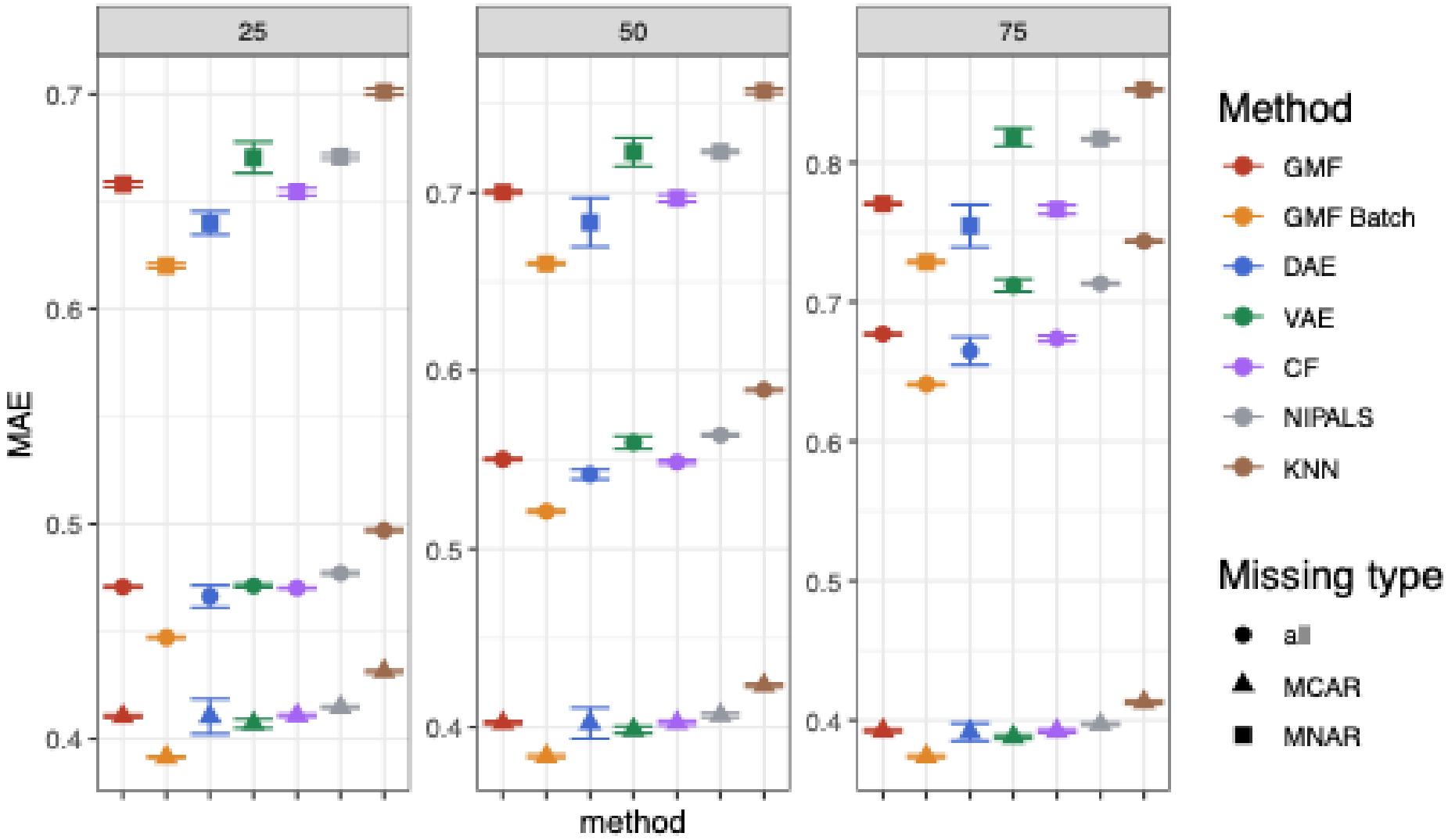
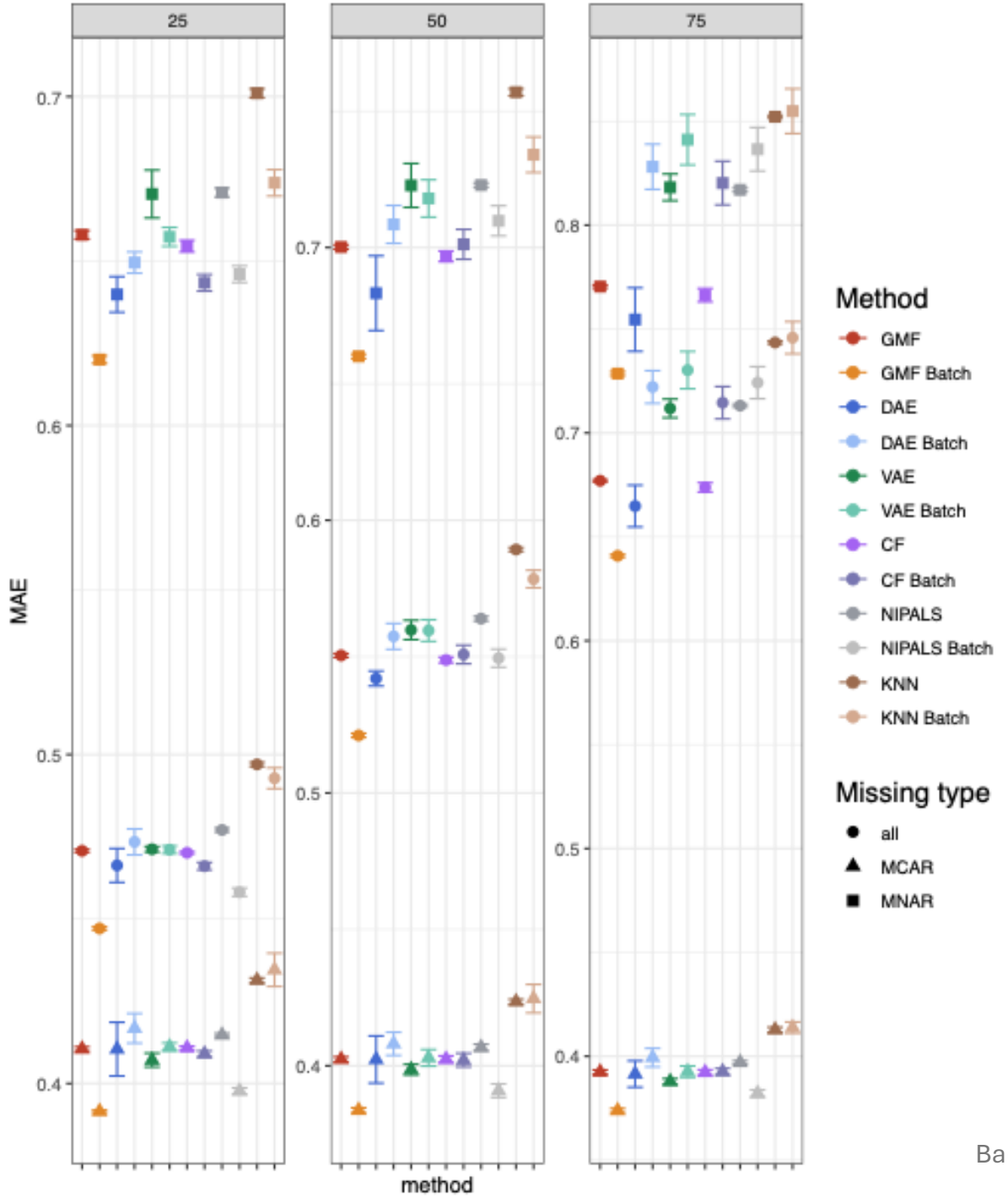
- Yeast background with 48 UPS proteins spiked in at 5 levels
- Analysed in triplicate in three labs → 45 samples
- 10105 peptides, 42.5% missing values
- Ionisation issues in Lab1 for samples A-C



Imputation: Missing Non-spiked Peptides



Imputation: Leduc



Imputation: Leduc